College Physics 2

Capacitor Electrical Energy Storage and Dielectric

MPS Department | FEU Institute of Technology

College Physics 2

Parallel Plate Capacitor and Dielectric

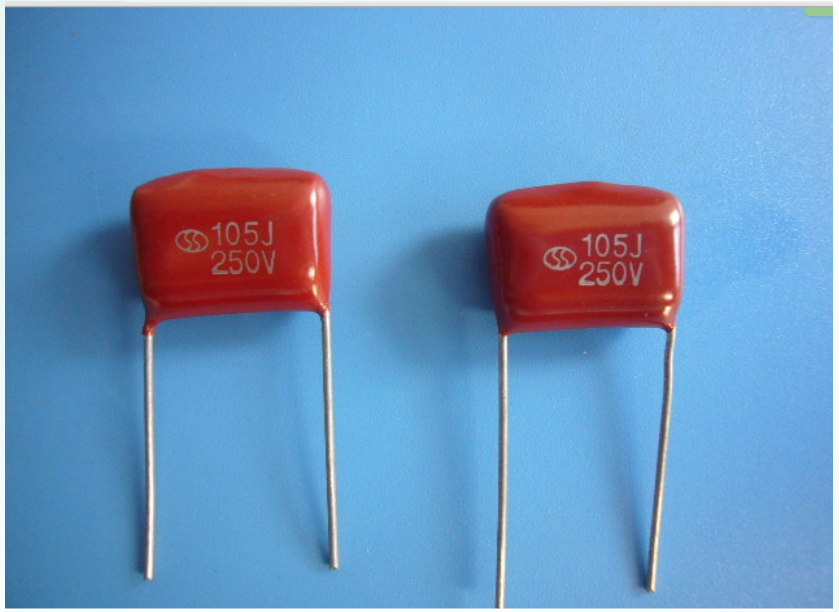
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OBJECTIVES

- Describe the nature of parallel-plate capacitors
- Discuss dielectric materials.
- Determine the effects of adding dielectrics in a capacitor.
- Calculate the amount of energy stored in a capacitor

CAPACITOR





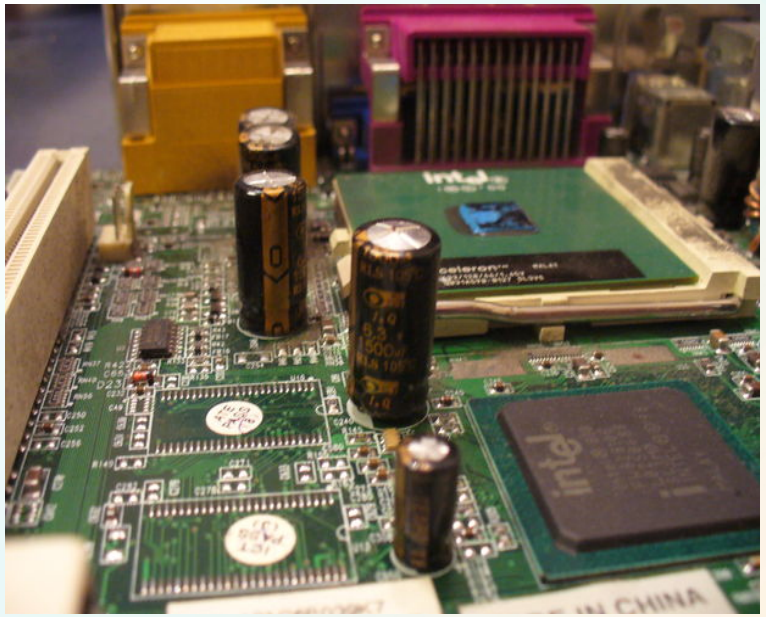


Figure 1 Types of Capacitors Source: Retrieved from.http://www.aliexpress.com and http://www.instructables.com

CAPACITOR AND ITS FUNCTION

A **capacitor** is a passive two terminal electrical component that stores potential energy in an electric field.



Figure 2. Assorted commercially available capacitors.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 793) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

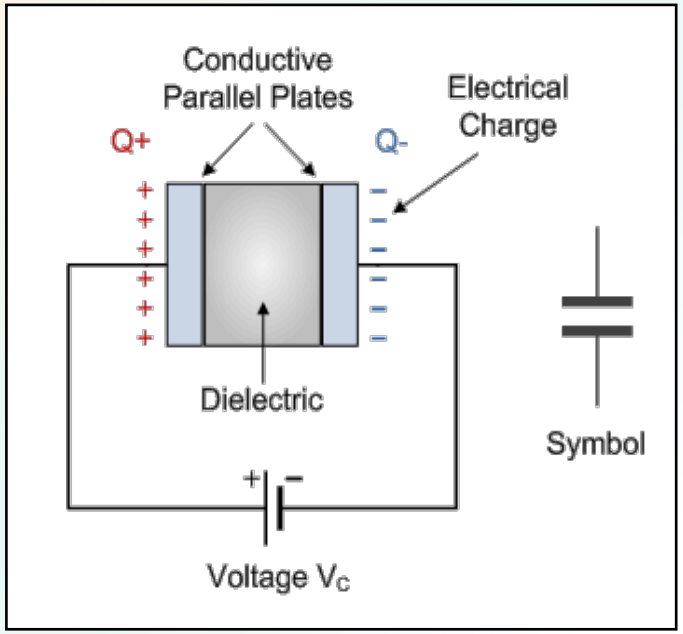


Figure 3. Conductive parallel plates. From.*https://www.electronics-tutorials.ws/wp-content/uploads/2018/05/capacitor-cap1.gif*

It consists of two **conductors** separated by a **dielectric**.

It is **charged when connected to a voltage source** and **discharged when connected to a load.**

DIELECTRICS

A **dielectric** is a non-conducting material between the conducting plates of a capacitor.

FUNCTIONS OF A DIELECTRIC

- allows very small separation between conducting plates w/o actual contact
- increases the maximum possible potential difference between the capacitor plates
- increases the capacitance of the capacitor

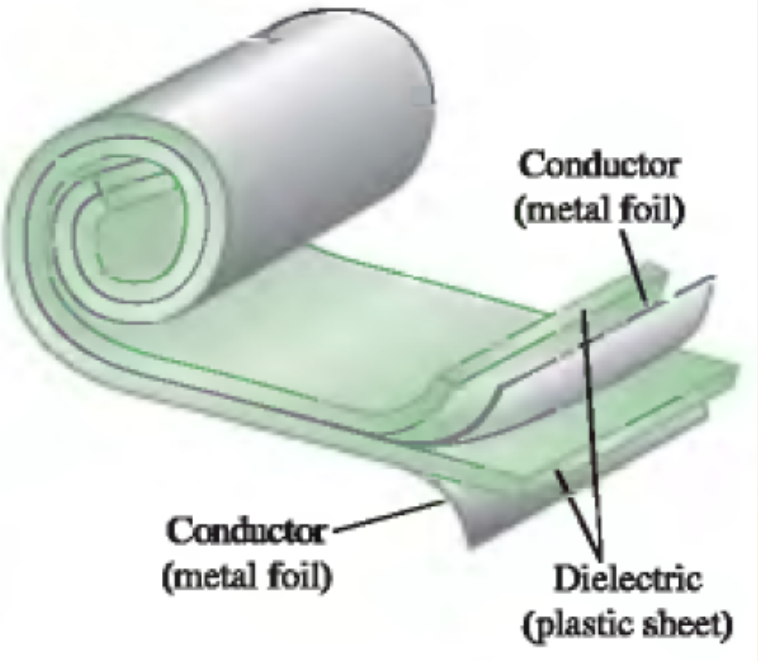


Figure 4. A common type of capacitor uses dielectric sheets to separate the conductors.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 800) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

CAPACITANCE

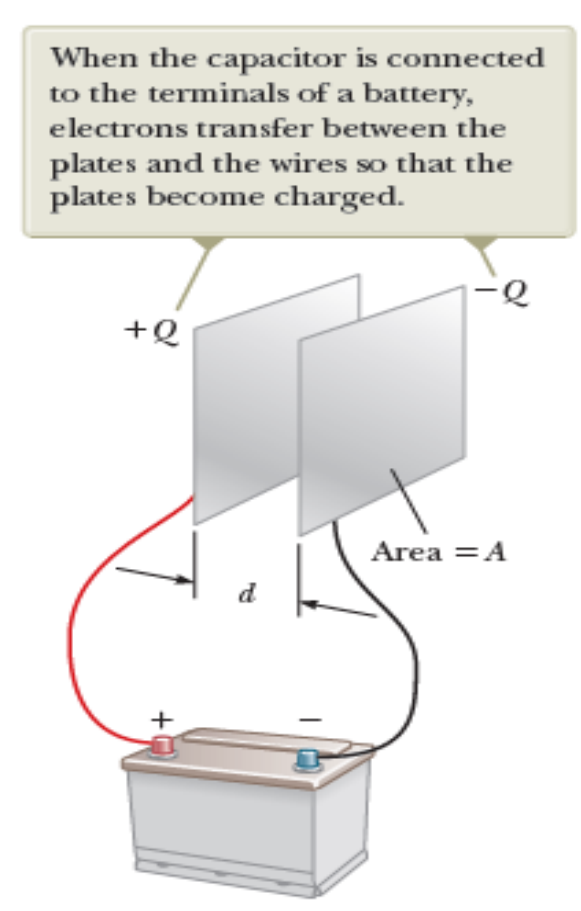


Figure 5. Parallel plate capacitor . From. *Physics for Scientists and Engineers with Modern Physics*(pp 778) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

- the plates have charges of equal magnitude but opposite signs $\boldsymbol{Q}$ and $\boldsymbol{-Q}$
- there's uniform electric field $\boldsymbol{E}$ between the plates
- there's potential difference $\boldsymbol{V_{ab}}$ between the plates

CAPACITANCE

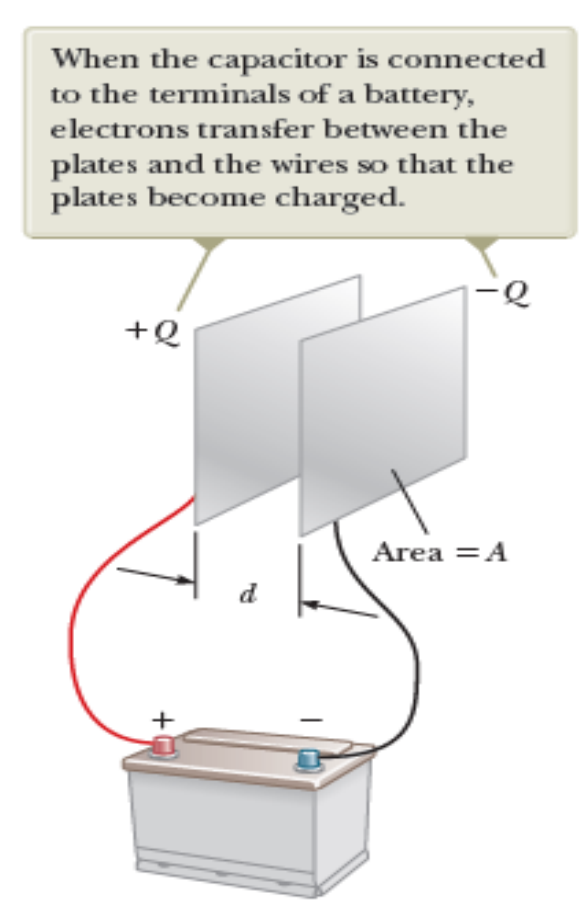


Figure 5. Parallel plate capacitor . From. *Physics for Scientists and Engineers with Modern Physics*(pp 778) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

The amount of charge that must be stored in the conductors to produce a certain potential difference is called the **capacitance** and is given by:

$$C = \frac{Q}{V_{ab}}$$

where:

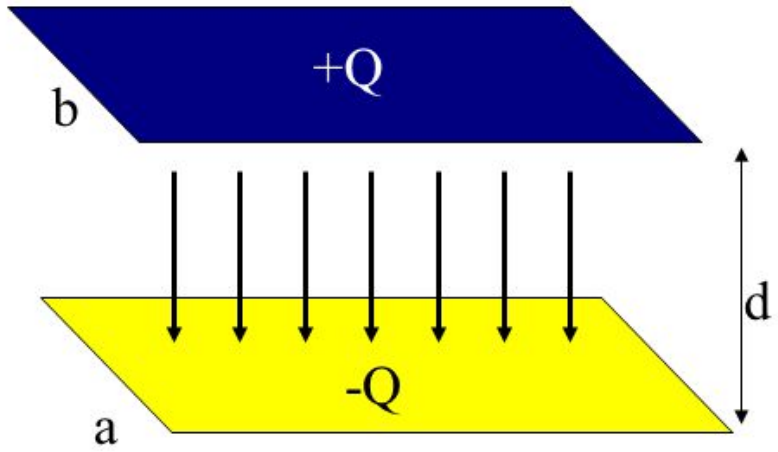
C is the capacitance in **farads (F)**

Q is the charge in **coulombs (C)**

V_{ab} is the potential difference in **volts (V)**

FACTORS AFFECTING CAPACITANCE

C is directly proportional with A



C is inversely proportional with d

For a parallel-plate capacitor consists of two plates each with area $\boldsymbol{A}$ and separated by a distance $\boldsymbol{d}$, the capacitance is given by:

$$\boldsymbol{C = \epsilon_0 \frac{A}{d}}$$

where:

$\boldsymbol{C}$ is the capacitance in **farads (F)**

$\boldsymbol{A}$ is the area of the plates in square meters $\mathbf{m^2}$

$\boldsymbol{d}$ is the separation distance of the plates in **meters (m)**

$\boldsymbol{\epsilon_0 = 8.85 \times 10^{-12}\ F/m}$

CAPACITANCE

Conceptual Question

Many computer keyboard buttons are constructed of capacitors as shown in the figure. When a key is pushed down, the soft insulator between the movable plate and the fixed plate is compressed. What happens to the capacitance when the key is pressed?

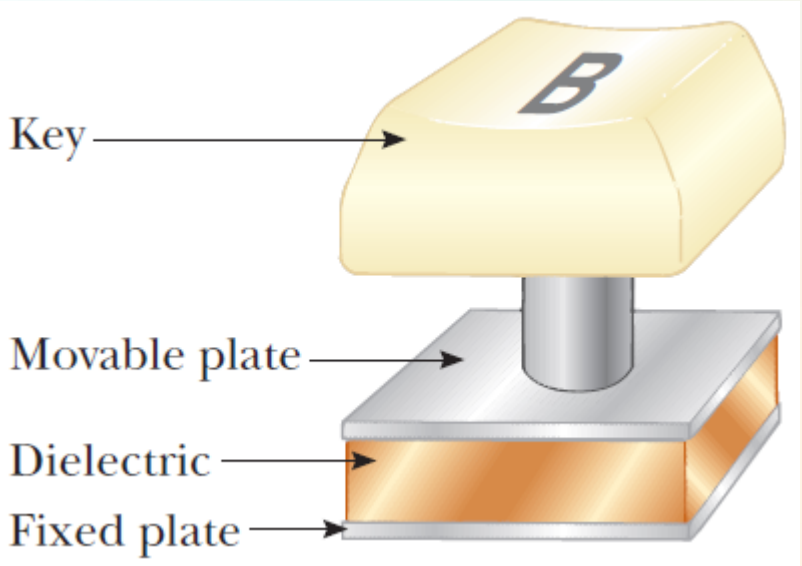
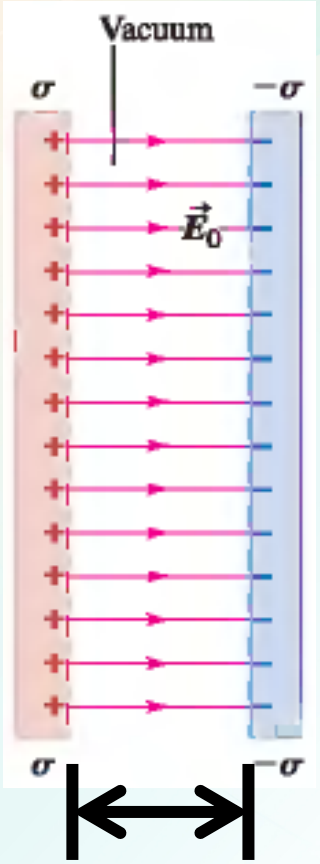


Figure 7. Computer Keyboard Button. From. *Physics for Scientists and Engineers with Modern Physics*(pp 780) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

***Ans:* It increases.**

Parallel Plate Capacitor



A

$$E = \frac{\sigma}{\varepsilon_o} \longrightarrow \sigma = \frac{Q}{A} \longrightarrow E = \frac{Q}{\varepsilon_0 A}$$

$$V_{ab} = Ed \longrightarrow V_{ab} = \frac{Qd}{\varepsilon_o A} \longrightarrow C = \frac{Q}{V_{ab}}$$

$$C = \varepsilon_o \frac{A}{d}$$

Depends only on the geometry

- ↑A, ↑ C
- ↑d, ↓C

- ↓ A, ↓ C
- ↓ d, ↑ C

Figure 8. Electric Field Lines with Vacuum between Plates .. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 801) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Values of Dielectric Constant at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310

Table 1. Values of Dielectric Constant.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 801) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Parallel Plate Capacitor

for other Dielectric Material

$$C = k\,\epsilon_0 \frac{A}{d}$$

where: k = dielectric constant

Sample Problems

1. A parallel plate capacitor has a capacitance of 1.0 F. If the plates are 1.0mm apart, what is the area of the plates?

ANSWER: 1.1×10^8 m^2

2. The plates of a parallel plate capacitor in vacuum are 5.0mm apart and 2.0m^2 in area. A potential difference of 10,000 V is applied across the capacitor. Compute (a) the capacitance (b) the charge on each plate; and (c) the magnitude of electric field

ANSWER: (a) 0.00354μF (b) 35.4μC (c) 2.0×10^6 N/C

ENERGY STORAGE IN CAPACITORS

The electric potential energy stored in a charged capacitor is just equal to the amount of work required to charge it. It is given by:

$$U = \frac{1}{2}CV_{ab}^2 = \frac{1}{2}QV_{ab}$$

where:

U is the electric potential energy in **joules (J)**

C is the capacitance in **farads (F)**

Q is the charge of the capacitor in **coulombs (C)**

V is the potential difference (voltage) in **volts (V)**

Sample Problems

3. A 5.80 μF, parallel-plate, air capacitor has a plate separation of 5.00 mm and is charged to a potential difference of 4.00×10^2 V. Calculate the energy stored in the capacitor.

ANSWER: 0.464 J

4. How much charge does a battery have to supply to a 5.0 μF capacitor to create a potential difference of 1.5 V across its plates?

ANSWER: 7.5 μC

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://www.electronics-tutorials.ws/wp-content/uploads/2018/05/capacitor-cap1.gif

College Physics 2

Capacitors in Series and Parallel

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OBJECTIVES

- Analyze capacitors connected in a network
- Calculate the total capacitance of capacitors in a network

CIRCUIT SYMBOLS

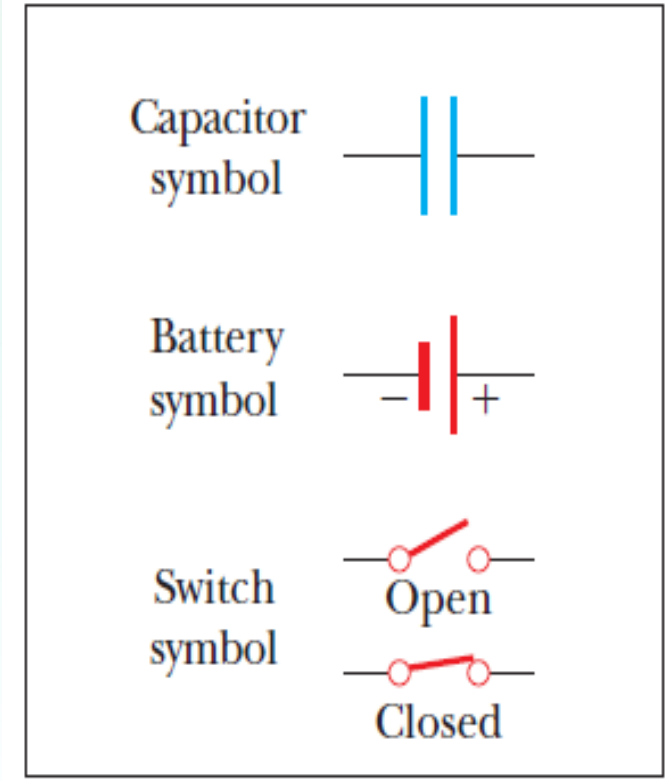


Figure 9 Computer Keyboard Button. From. *Physics for Scientists and Engineers with Modern Physics*(pp 782) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

Circuit symbols for capacitors, batteries, and switches.

CAPACITORS IN SERIES AND PARALLEL CONNECTION

For capacitors in series:

$$Q_T = Q_1 = Q_2 = Q_3$$

$$V_{ab} = V_1 + V_2 + V_3$$

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

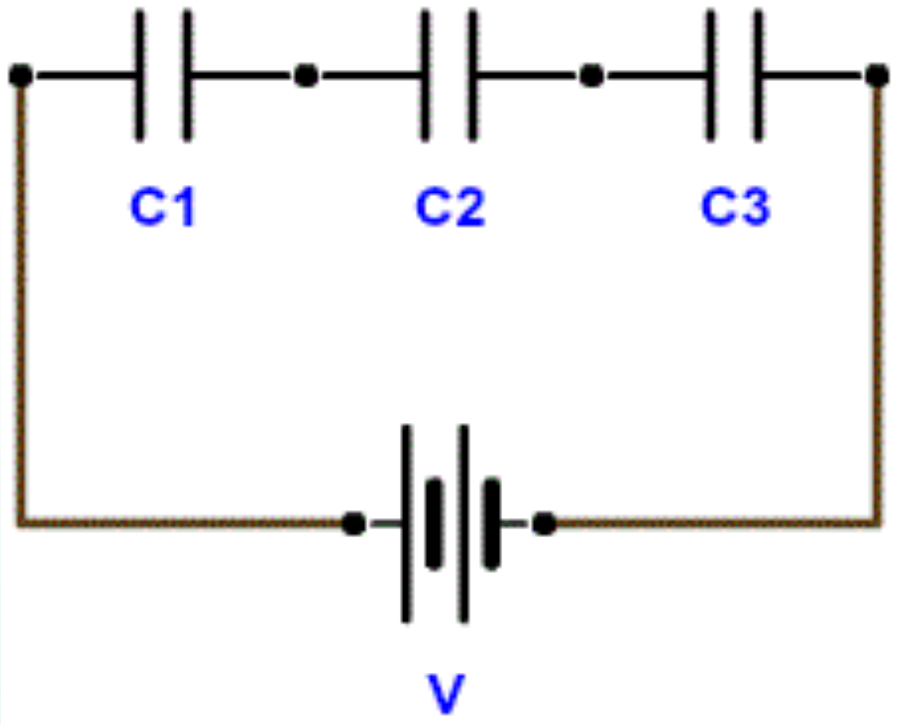


Figure 10. Capacitors in series. From. *https://cdn.shortpixel.ai/client/q_glossy,ret_img,w_228,h_183/https://electronicspani.com/wp-content/uploads/2014/11/Capacitors-in-Series.gif*

CAPACITORS IN SERIES AND PARALLEL CONNECTION

For capacitors in parallel:

$$V_{ab} = V_1 = V_2 = V_3$$

$$Q_T = Q_1 + Q_2 + Q_3$$

$$C_T = C_1 + C_2 + C_3$$

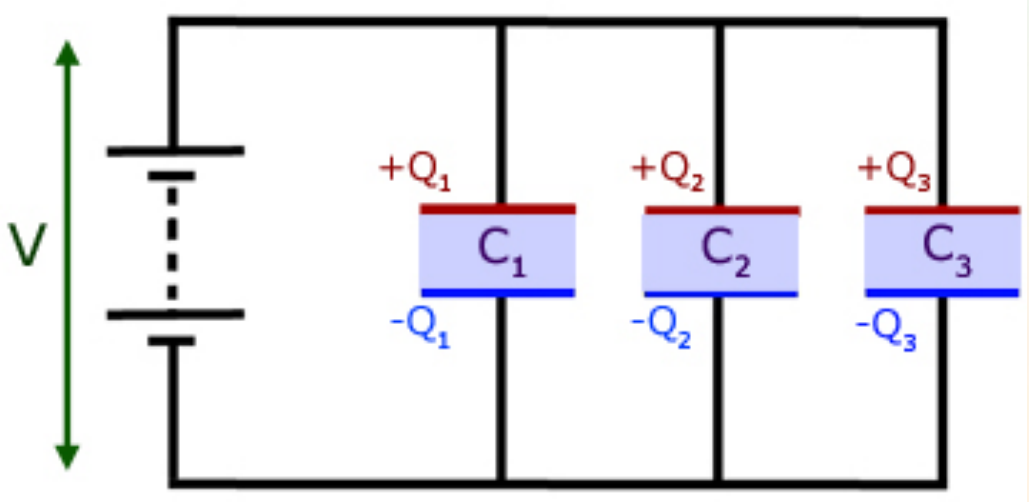


Figure11. Capacitors in parallel..*From.https://tse3.mm.bing.net/th?id=OIP.9OfBunwrgzrkznVZBk1DCQAAAA&pid=Api&P=0&w=322&h=158*

CAPACITORS IN SERIES AND PARALLEL CONNECTION

Conceptual Question

You want to connect a $4\,\mu F$ capacitor and an $8\,\mu F$ capacitor. What type of connection will the $4\,\mu F$ capacitor have a greater potential difference across it than the $8\,\mu F$ capacitor?

Ans: **series connection**

Sample Problems

1. Two capacitors, $C_1 = 6.0\ \mu F$ and $C_2 = 3.0\ \mu F$ are connected to a $18.0\ V$ battery. Find the total capacitance and find the charge and potential difference for each capacitor when the two capacitors are connected (a) in series and (b) in parallel

   ANSWER: (a) 2 µF (b) 9 µF

2. Two capacitors, $C_1 = 5.0\ \mu F$ and $C_2 = 12.0\ \mu F$ are connected in parallel to a $9.0\ V$ battery. Find (a) the total capacitance and (b) the potential difference across each capacitor.

   ANSWER: (a) 17 µF (b) 9.0 V

Sample Problem

Find the (a) total capacitance, (b) potential difference and (c) charge across each capacitor for the combination shown in the figure. Potential difference between pt. a and b is V_{ab} = 28 V.

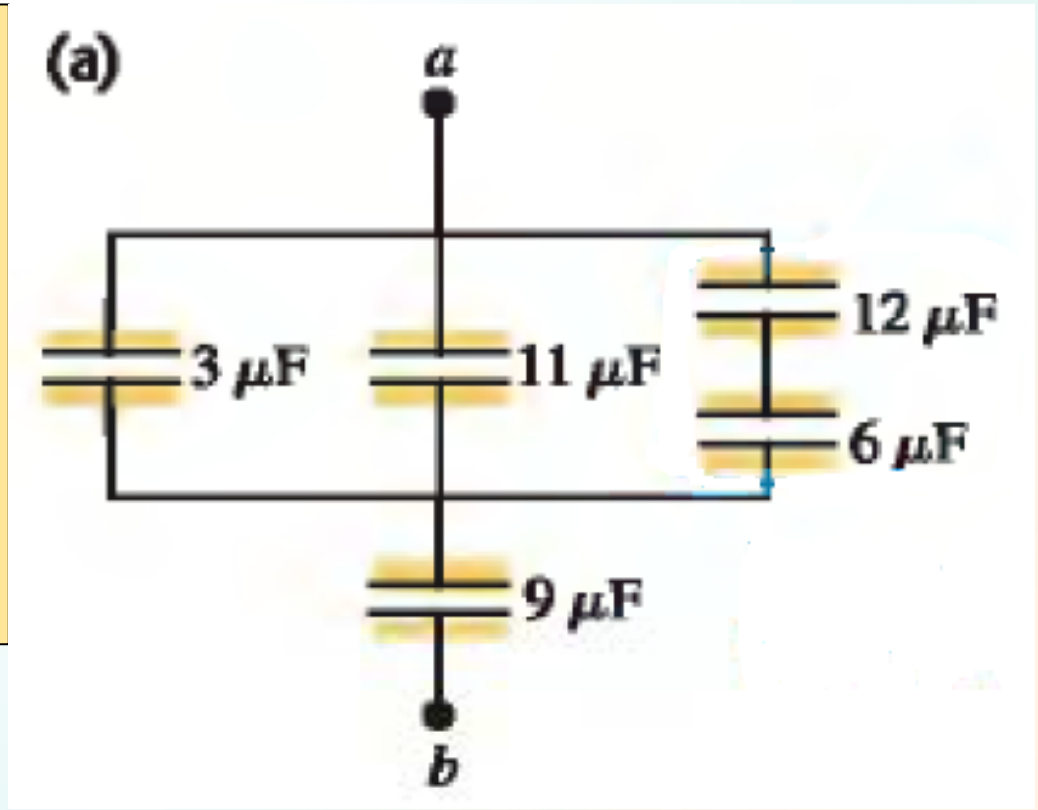


Figure 12. Capacitors in series and parallel.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 796) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

APPLICATIONS OF CAPACITORS



Figure 13. Application of capacitors. From. h*ttps://magazine.odroid.com/article/odroid-c2-power-off-wake-using-simple-gpio-button/*



Figure 14. TV Remote control. From.https://tse3.mm.bing.net/th?id=OIP.J5mtsttLmDjNQI17QLBHpwHaHa&pid=Api&P=0&w=300&h=300



Figure 15. Camera with flash. From. *https://c.pxhere.com/photos/68/0e/camera_digital_equipment_female_girl_isolated_lens_people-1260303.jpg!d*

protection when there are power surges

provides charge at a greater rate

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://cdn.shortpixel.ai/client/q_glossy,ret_img,w_228,h_183/https://electronicspani.com/wp-content/uploads/2014/11/Capacitors-in-Series.gif

https://tse3.mm.bing.net/th?id=OIP.9OfBunwrgzrkznVZBk1DCQAAAA&pid=Api&P=0&w=322&h=158

https://tse3.mm.bing.net/th?id=OIP.J5mtsttLmDjNQI17QLBHpwHaHa&pid=Api&P=0&w=300&h=300

https://c.pxhere.com/photos/68/0e/camera_digital_equipment_female_girl_isolated_lens_people-1260303.jpg!d